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VAPI: Verifiable Controller Input Provenance with Physics-Backed Liveness for Competitive Gaming

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Abstract

Part 1 — Provenance. VAPI provides a cryptographically verifiable evidence rail for controller input. Each gaming session produces a signed, hash-chained stream of 228-byte evidence records whose origin (ECDSA-P256 signature), ordering (monotonic counter + hash linkage), and integrity (hash-chain verification) can be confirmed by any third party on a public blockchain without access to the original device.

Part 2 — Physics. Software-only injection is made empirically infeasible by a nine-level Physical Input Trust Layer (PITL) that binds committed evidence to physics-coupled controller signals — IMU gravity baseline, IMU-button causal latency, stick-IMU temporal cross-correlation, biometric kinematic fingerprinting (11 features, Mahalanobis distance), temporal rhythm analysis, and active haptic challenge-response using the DualShock Edge's motorized adaptive triggers. Live hardware validation on a DualShock Edge CFI-ZCP1 confirms a 14,000× injection detection margin.

Part 3 — Status. The prototype spans ~220 files with ~1,289 automated tests (865 bridge, 354 contract, 28 SDK, 14 E2E, 28 hardware). Thirteen contracts are deployed on IoTeX testnet. Living calibration (Mode 6, Phase 38) autonomously evolves L4 thresholds from verified session data every 6 hours using exponential decay weighting. All PITL thresholds are empirically calibrated from N=69

real sessions across 3 distinct players. The primary current limitation is single-population calibration: L4 functions as a per-player anomaly detector rather than a cross-player identifier (inter-person separation ratio 0.362).

Keywords: proof of cognition, gaming anti-cheat, verifiable gaming intelligence, physical human controller input, PHCI certification, adaptive trigger attestation, federated threat detection, on-chain verification

ACM CCS Concepts: Security and privacy → Authentication; Computer systems organization → Embedded and cyber-physical systems; Computing methodologies → Multi-agent systems

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1. Introduction

Competitive online gaming operates without any cryptographic guarantee that the person pressing the buttons is a human. Game servers trust client-reported inputs; anti-cheat software (Easy Anti-Cheat, BattlEye, Vanguard) detects known cheat signatures at the kernel level but cannot prove human operation — only the absence of *known* automated tools. A sufficiently sophisticated bot can pass every existing client-side check by injecting inputs directly into the HID driver stack, bypassing behavioral analysis entirely. Professional esports competitions worth millions of dollars in prize money have no cryptographic defense against this class of attack.

We identify the root gap as the **human-controller attestation problem**: given a player who claims to have performed a gaming session on a physical controller, how can a tournament operator — with no access to the player's hardware — verify the provenance and physical plausibility of a player's controller input stream?

This paper makes five contributions toward solving this problem:

1. Proof of Autonomous Cognition (PoAC). A 228-byte cryptographic evidence rail that chains sensor commitments, model manifests, world-model hashes, and inference results into a tamper-evident, hash-linked sequence. Each record attests not just to *what* a device sensed but to *why* it acted — capturing the decision context through a commitment to accumulated agent state (§4). (PoAC proves commitment to cognitive context, not correctness of cognition.)

2. Physical Input Trust Layer (PITL). A nine-level physics-backed liveness detection stack combining hard structural checks (HID-XInput pipeline discrepancy, PoAC chain integrity) with adaptive behavioral analysis (biometric Mahalanobis fingerprinting, temporal rhythm analysis, behavioral archaeology, network correlation) and active physical challenge-response (randomized adaptive trigger resistance profiling with motor-response curve classification). Five of the nine layers exploit signals that cannot be replicated by software injection (§7.5.1, §7.5.2). Section §7.5.2.1 derives game genre certification requirements: L4 biometric activation depends on controller usage patterns specific to game genre, producing four deployment tiers (FULL/STANDARD/LIMITED/NOT RECOMMENDED). NCAA Football 26 is classified LIMITED CERTIFICATION (3 active features; L4 operates as intra-player anomaly detector only, not inter-player identifier).

3. PHG Humanity Credential. A soulbound, non-transferable on-chain credential (ERC-5192-inspired, `locked()=true`) whose validity reflects continuous behavioral surveillance. Credentials are provisionally suspended when a device accumulates consecutive critical behavioral windows and automatically reinstated when it clears — making the credential a *living proof* of ongoing trustworthy behavior, not a one-time certification (§7.5.4).

4. Adaptive Detection Feedback Loop. An anti-cheat system where retrospective behavioral memory directly drives forward detection policy: devices labeled `critical` have their L4 Mahalanobis detection threshold tightened by 30% for subsequent sessions. The loop is bounded (minimum multiplier floor 0.5×), reversible (label changes auto-restore the threshold), and cryptographically bounded (the 228-byte PoAC wire format is unchanged) (§7.5.5). Phase 38 adds Mode 6 living calibration: thresholds evolve autonomously from verified session data every 6 hours using exponential decay weighting (α=0.95), bounded to ±15% per cycle.

5. Federated Cross-Instance Threat Correlation. A privacy-preserving federation protocol that exchanges cluster fingerprints (16-char SHA-256 hashes, non-reversible) between independent bridge instances, enabling detection of bot farms that deliberately distribute devices across shards to stay below each instance's local threshold (§7.5.6).

Together these contributions form **VAPI** (Verified Autonomous Physical Intelligence). The primary certified device is the **DualShock Edge** (Sony CFI-ZCP1), whose motorized adaptive trigger surface creates a detection surface that software cannot cross. The primary blockchain is **IoTeX L1**, leveraging its native P256 precompile for gas-efficient ECDSA verification. A secondary IoTeX Pebble Tracker integration (§7.5.9) validates protocol extensibility to DePIN sensor domains — the same 228-byte wire format operates unchanged.

The remainder of this paper is organized as follows. §2 surveys related work. §3 formalizes the system model. §4 presents the PoAC protocol. §5 describes the agent architecture. §6 introduces the DePIN economic layer. §7 details the implementation, with §7.5 presenting the complete DualShock Edge anti-cheat subsystem organized by conceptual layer rather than development chronology. §8 presents evaluation results. §9 analyzes security and threat models. §10 discusses limitations and future directions. §11 concludes.

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2. Background and Related Work

2.1 Gaming Anti-Cheat: The Detection Gap

Existing anti-cheat systems (Easy Anti-Cheat, BattlEye, Vanguard) operate as kernel drivers that scan for known cheat signatures in process memory and loaded modules. They detect *what software is running* but cannot detect *whether inputs are human-generated*. A driver-level HID injection attack — spoofing the USB report stream that the game receives — defeats all signature-based detection because it operates below the HID driver abstraction layer and produces reports indistinguishable from a real controller.

The literature on game bot detection focuses on behavioral analysis of input sequences [19] and statistical anomaly detection over timing distributions [20]. VAPI complements these approaches by adding a cryptographic layer: behavioral signals are committed and chain-linked, making their provenance verifiable to a third party who was not present during the session.

2.2 Decentralized Physical Infrastructure Networks

DePIN protocols incentivize the deployment and operation of physical infrastructure through token rewards. Helium [2] pioneered Proof of Coverage for wireless networks. Hivemapper [3] rewards dashcam operators for contributing street-level imagery. These systems verify *presence* or *data quality* but none verify the *cognitive process* by which a device interprets its environment. VAPI's DePIN

extensibility (§7.5.9) demonstrates that PoAC fills this gap for environmental monitoring devices, but the primary contribution of this paper is gaming anti-cheat.

2.3 Trusted Execution and Remote Attestation

ARM TrustZone [6] and Intel SGX [9] verify *what code is running* but not *what the code perceives or decides*. PoAC complements platform attestation by extending the trust chain from code identity to cognitive content — committing actual sensor readings, model weights, accumulated state, and inference outputs.

2.4 Verifiable Computation and zkML

Verifiable computation systems [11] allow proving correct execution of arbitrary programs. Recent zkML efforts [12, 13] apply zero-knowledge proofs to ML inference. While theoretically powerful, zkML faces severe practical barriers on microcontrollers: proof generation requires minutes for small models on desktop hardware. PoAC adopts a pragmatic alternative: rather than proving *correct execution* of inference (zkML), we prove *commitment to the complete cognitive context*, trading zero-knowledge guarantees for real-time feasibility. VAPI does integrate a Groth16 ZK circuit for PITL session proofs (§7.5.3), but at the session level (one proof per gaming session, not per cognition cycle).

2.5 Autonomous Agent Architectures

The BDI model [16] decomposes agent reasoning into beliefs, desires, and intentions. Subsumption architectures [17] layer reactive behaviors with priority-based arbitration. VAPI's three-layer architecture (reflexive/deliberative/strategic) draws on both: the world model encodes beliefs, bounty evaluation encodes desires, and PoAC records encode intentions. Critically, VAPI operates entirely on-device with no cloud dependency for core cognition.

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3. System Model and Definitions

Definition 1 (Embodied Agent). An embodied agent $\mathcal{A}$ is a tuple $(\mathcal{S}, \mathcal{M}, \mathcal{W}, \mathcal{D}, \mathcal{K})$ where $\mathcal{S}$ is a sensor suite, $\mathcal{M}$ is an inference model, $\mathcal{W}$ is a world model (accumulated state), $\mathcal{D}$ is a decision function, and $\mathcal{K}$ is a signing keypair.

Definition 2 (Cognition Cycle). A single cognition cycle c_i at time t_i consists of: 1. **Perception**: $s_i \leftarrow \text{sense}(\mathcal{S}, t_i)$ 2. **Commitment**: $h_s \leftarrow H(s_i)$, $h_m \leftarrow H(\mathcal{M})$, $h_w \leftarrow H(\mathcal{W})$ 3. **Inference**: $(y_i, p_i) \leftarrow \mathcal{M}(s_i)$ 4. **Decision**: $a_i \leftarrow \mathcal{D}(y_i, p_i, \mathcal{W})$ 5. **Attestation**: $\rho_i \leftarrow \text{sign}(\mathcal{K}, [h_{i-1} \| h_s \| h_m \| h_w \| y_i \| a_i \| \text{ctx}_i])$ 6. **Update**: $\mathcal{W} \leftarrow \mathcal{W} \cup \{(s_i, y_i, a_i)\}$

Definition 3 (PoAC Chain). A PoAC chain for device d is an ordered sequence $\langle \rho_0, \rho_1, \dots, \rho_n \rangle$ where $\rho_i.\text{prev_hash} = \text{SHA-256}(\rho_{i-1}.\text{body}_{164})$ and $\rho_i.\text{ctr} > \rho_{i-1}.\text{ctr}$, forming a hash-linked, monotonically-ordered evidence log. Note: the chain link hash uses the 164-byte signed body only; the 64-byte ECDSA signature is excluded from the hash to match `PoACVerifier.sol`.

Definition 4 (PHG Credential State). A device's PHG credential is in one of three states: `stable` (actively accumulating humanity evidence), `suspended` (credential provisionally revoked due to consecutive critical behavioral windows), or `cleared` (previously suspended, now exhibiting clean behavior). Transitions are driven by `InsightSynthesizer` (§7.5.5) and recorded on-chain.

Threat Model. We consider an adversary who: (T1) fabricates PoAC records without the device key; (T2) replays valid records out of order; (T3) selectively omits records to hide unfavorable evidence; (T4) injects synthetic sensor data to a legitimate device; (T5) claims rewards for work not performed; (T6) executes a warm-up attack — gradually training a bot to produce human-like behavioral signals over many sessions; (T7) distributes a bot farm across multiple bridge shards to stay below each instance's local detection threshold. We assume the device hardware is not physically compromised and that the blockchain provides standard finality guarantees.

What VAPI Proves vs What It Infers

The following table distinguishes cryptographic guarantees from empirical inferences and explicit non-claims. Understanding this distinction is essential to honest evaluation of the system.

Table: What VAPI Proves vs What It Infers

Property	Guarantee Type	Mechanism
Record origin (signed by registered key)	Cryptographic	ECDSA-P256 signature, key in CryptoCell-310 secure element
Record ordering (no reordering/insertion)	Cryptographic	Monotonic counter + hash-chain linkage
Replay resistance	Cryptographic	Monotonic counter + ZK nullifier/epoch
Chain integrity (omission detectable)	Cryptographic (verifiable gaps)	Missing counter values visible to any chain verifier
Physical human operation of controller	Empirical inference	PITL L0–L6 physics-coupled signals
Software injection absence	Empirical inference	IMU noise floor, gravity, causal coupling, temporal rhythm
Player identity across sessions	NOT currently proven — game-genre dependent	Separation ratio 0.362 for NCAA CFB 26 (3 active features); FPS/Racing genres expected > 2.0 ratio with 9–10 active features. See §7.5.2.1.
Biometric pipeline correctness (end-to-end)	NOT proven on-chain	ZK proof binds feature commitment and nullifier; raw→feature transformation is trusted

Property	Guarantee Type	Mechanism
Inference code on-chain	Enforced via ZK circuit C2 (Phase 41)	`pub[2]=inferenceCode` in PITLSessionRegistry.sol; circuit constraint `inferenceResult` □ `[40,42]` makes cheat-code proofs ungenerable. Inference also committed off-chain in bridge SQLite.
Bridge execution honesty	Trust assumption (constrained)	Withholding detectable via chain gaps; computation constrained by ZK circuit; enforcement mediated

---

4. The Proof of Autonomous Cognition Protocol

4.1 Record Structure

Each PoAC record is a fixed-size 228-byte structure: a 164-byte signed body and a 64-byte ECDSA-P256 signature. The fixed-size design eliminates parsing ambiguity, enables zero-copy deserialization, and fits within a single NB-IoT uplink frame.

Table 1: PoAC Record Wire Format (228 bytes, FROZEN)

Offset	Field	Size	Description
`0x00`	`prev_poac_hash`	32 B	SHA-256 of previous record's 164-byte body (genesis: `0x00...0`)
`0x20`	`sensor_commitment`	32 B	$H(\text{raw_sensor_buffer})$
`0x40`	`model_manifest_hash`	32 B	$H(\text{weights}\|\text{version}\|\text{arch_id})$
`0x60`	`world_model_hash`	32 B	$H(\mathcal{W})$ — agent state *before* update
`0x80`	`inference_result`	1 B	Encoded classification output
`0x81`	`action_code`	1 B	Agent action
`0x82`	`confidence`	1 B	Model confidence $\in [0, 255]$
`0x83`	`battery_pct`	1 B	Remaining energy $\in [0, 100]$
`0x84`	`monotonic_ctr`	4 B	Strictly increasing counter (big-endian)
`0x88`	`timestamp_ms`	8 B	Unix epoch milliseconds
`0x90`	`latitude`	8 B	WGS84 latitude
`0x98`	`longitude`	8 B	WGS84 longitude
`0xA0`	`bounty_id`	4 B	On-chain bounty reference
`0xA4`	`signature`	64 B	ECDSA-P256: $r\|s$

Hashing invariants (CRITICAL): - `record_hash` = SHA-256(raw[0:164]) — the 164-byte signed body; this value is stored as `prev_poac_hash` in the *next* record

and as `lastRecordHash` in `PoACVerifier.sol` - `chain_hash` = SHA-256(raw[0:228]) — the full-record hash including signature; used as an off-chain convenience for indexing and de-duplication only; NOT used for chain linkage - These are different values. Using `chain_hash` as a chain link produces verification failures against `PoACVerifier.sol`.

The four 32-byte hash commitments (offsets `0x00–0x7F`) distinguish PoAC from simple signed telemetry: they capture not just *what* was observed but the complete cognitive context of *why* a particular action was chosen.

4.2 Chain Integrity

The `PoACVerifier` contract enforces four properties per submitted record: - `submission.monotonicCtr > chainState.lastCounter` (ordering) - `submission.prevPoACHash == chainState.lastRecordHash` (linkage) - `|submission.timestampMs - block.timestamp × 1000| ≤ maxTimestampSkew` (freshness) - Valid ECDSA-P256 signature over the 164-byte body (authenticity)

4.3 PITL Inference Codes

Code	Name	Layer	Type
`0x20`	NOMINAL	—	Normal human
`0x28`	DRIVER_INJECT	L2	Hard cheat — blocks PHG, SkillOracle, tournament
`0x29`	WALLHACK_PREAIM	L3	Hard cheat
`0x2A`	AIMBOT_BEHAVIORAL	L3	Hard cheat
`0x2B`	TEMPORAL_ANOMALY	L5	Advisory — committed but not blocking
`0x30`	BIOMETRIC_ANOMALY	L4	Advisory — committed but not blocking
`0x31`	IMU_BUTTON_DECOUPLED	L2B	Advisory — IMU precursor absent before button press
`0x32`	STICK_IMU_DECOUPLED	L2C	Advisory — stick-IMU temporal correlation absent

Hard codes `{0x28, 0x29, 0x2A}` are rejected by `TeamProofAggregator` and trigger a −200-point `SkillOracle` penalty. Advisory codes `{0x2B, 0x30}` accumulate as on-chain statistical evidence without directly affecting rating or eligibility.

5. Agent Architecture

VAPI implements a three-layer cognitive architecture adapted for resource-constrained gaming controller hardware.

5.1 Layer 1: Reflexive (Period: 1 ms at 1 kHz gaming input rate)

The reflexive layer executes the core sense-infer-attest loop. For the DualShock Edge subsystem: capture a 50-byte `InputSnapshot`, extract a 30-dimensional feature vector (stick positions, velocities, accelerations, trigger values, button timing statistics, IMU features, touchpad features, temporal features), run the INT8-quantized classifier (30→64→32→6), generate a PoAC record, and submit to the batch queue.

Detection thresholds (invariant to adaptive policy): - Hard cheat: confidence $\geq$ 180/255 ($≈$70%) triggers `CHEAT_ALERT` state - Resolution: 10 consecutive clean windows required to exit `CHEAT_ALERT`

5.2 Layer 2: Deliberative (Period: 60 s)

Trend analysis, EWC world model updates, behavioral archaeology signals, and PHG checkpoint commit decisions. If behavioral scores (warmup, burst) exceed thresholds, the PHG score delta is multiplied by `max(0.0, 1.0 - warmup×0.8 - burst×0.5)` before on-chain submission.

5.3 Layer 3: Strategic (Period: 5 min)

Bridge sync, session proof generation, ZK PITL proof submission. The autonomy guard prevents external systems from disabling PoAC generation.

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6. DePIN Economic Layer

The DePIN layer demonstrates PoAC extensibility to non-gaming sensor domains. Devices autonomously evaluate, accept, and preempt environmental monitoring bounties using a greedy knapsack optimizer (§6.3 of the original paper). Each economic decision is itself a PoAC-attested record. This layer is not the primary contribution of this paper; see Appendix A for complete details.

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7. System Implementation

7.1 Overview

VAPI spans four implementation layers: firmware (C, Zephyr RTOS and ESP-IDF), smart contracts (Solidity, Hardhat, IoTeX), a Python asyncio bridge service, and a DualShock Edge controller anti-cheat subsystem. The prototype comprises ~220 files (~1,289 automated tests total, ~1,261 in CI excluding hardware).

Table 2: Implementation Component Summary

Component	Language	Key Dependencies
Firmware	C (Zephyr)	nRF Connect SDK, PSA Crypto, CryptoCell-310
Controller	C (ESP-IDF) / Python	ESP-NN, TFLite Micro, pydualsense
Contracts	Solidity	OpenZeppelin, Hardhat, IoTeX P256 precompile
Bridge	Python	asyncio, Web3.py, aiomqtt, FastAPI, anthropic
SDK	Python + C99	pytest, VAPISession, VAPIVerifier

7.2 Smart Contract Stack

Core verification: - `PoACVerifier` — P256 signature verification via IoTeX precompile `0x0100`, batch verification up to 10 records, chain state enforcement - `TieredDeviceRegistry` — device identity, staking, reputation ($R = R_{\text{raw}} \times 10000/(R_{\text{raw}} + 1000)$ with diminishing returns)

Gaming intelligence: - `PHGRegistry` — PHG score checkpoints; `commitCheckpoint(deviceId, scoreDelta, count, biometricHash)` - `PHGCredential` — soulbound credential; `suspend(bytes32,bytes32,uint256)`, `reinstate(bytes32)`, `isActive(bytes32)` - `TournamentGateV3` — wraps V2's cumulative+velocity gates with `phgCredential.isActive()` as a third condition - `SkillOracle` — ELO-inspired on-chain skill rating; NOMINAL/SKILLED increment, cheat −200 penalty - `ProgressAttestation` — verifiable proof of improvement between two PoAC records - `TeamProofAggregator` — Merkle-root team attestation; rejects records with hard cheat codes - `PITLSessionRegistry` — ZK PITL session proofs; anti-replay via `usedNullifiers`. In production mode (`pitlVerifier ≠ address(0)`), the Groth16 proof cryptographically enforces feature commitment, humanity probability, nullifier uniqueness, and epoch binding. Known limitation: the inference code (`inferenceResult`) is passed as pub[2]=0 and not currently enforced on-chain; it is committed off-chain in the bridge SQLite store.

Federation: - `FederatedThreatRegistry` — immutable on-chain anchor for cross-confirmed cluster hashes

7.3 Bridge Service

The bridge is a Python asyncio application providing three transport ingestion channels (MQTT, CoAP, HTTP), a batch accumulator (bounded `asyncio.Queue(maxsize=1000)`), six intelligence background tasks, and two API sub-applications.

Background tasks: 1. `ChainReconciler` — confirms PHG checkpoint receipts every 30 s 2. `ProactiveMonitor` — bot-farm cluster + high-risk trajectory + eligibility surveillance every 60 s 3. `FederationBus` — peer cluster fingerprint exchange every 120 s 4. `InsightSynthesizer` — 6-mode longitudinal synthesis (Mode 6: living calibration) + detection policies every 6 h 5. `AlertRouter` — webhook dispatch for enforcement events every 30 s 6. `BridgeAgent` — LLM-powered operator intelligence (on-demand, not polled)

7.5 DualShock Edge: Proof of Human Gaming

7.5.1 The Adaptive Trigger Detection Surface

The DualShock Edge (CFI-ZCP1) is the production PHCI (Physical Human Controller Input) certified device. Its defining hardware feature is the motorized L2/R2 adaptive trigger system, generating programmable resistance force profiles at approximately 200 Hz.

Unlike every other controller sensor — stick axes, gyroscope, accelerometer, buttons — the trigger resistance dynamics emerge from a mechanical actuator whose behavior is governed by physical laws (spring constants, friction, thermal noise, electromagnetic hysteresis) that software cannot replicate or inject. A cheating player using software-injected inputs can reproduce stick trajectories and button timings, but cannot reproduce the micro-jitter, onset velocity profile, and grip-force ratio that a real human hand produces against a real adaptive trigger.

The DualShock Edge adaptive trigger surface is a physics-backed liveness detection surface that cannot be crossed by software-only injection.

Sensor commitment schema v2 (kinematic/haptic) commits into every PoAC record's 32-byte `sensor_commitment` field: - Left/right stick axes (4 × int16) - Trigger depression values (2 × uint8) - **Trigger resistance effect mode** (2 × uint8) — read from controller ADC, not writable from host HID stack - Gyroscope (3 × int16) - Accelerometer (3 × int16) - Timestamp (int64)

L6 challenge extension (conditional, +4 bytes → 52 bytes total). When L6 Active Physical Challenge-Response is enabled and a challenge is in flight, the sensor commitment preimage is extended with: challenge `profile_id` (uint8), 16-bit SHA-256 profile fingerprint (uint16), and `l6_score_int` = $\lfloor$p_human_L6 × 100$\rfloor$ (uint8). When L6 is disabled or no challenge is pending, the preimage remains 48 bytes — producing byte-identical commitments to any pre-L6 record. The extension is conditional and non-breaking.

Any software injection that fails to reproduce physical resistance dynamics produces a measurably different sensor commitment, breaking hash-chain consistency.

7.5.2 Physical Input Trust Layer

VAPI implements a nine-level detection stack. Each level is independent; detections at any level produce PITL inference codes committed into the PoAC record. L2B and L2C are independent advisory oracles within the L2 detection band, exploiting temporal coupling physics absent from software injection.

Table 3: PITL Architecture (nine-level)

Layer	Module	Code	Type	Signal
L0	Physical presence	—	Structural	Controller must be HID-connected, live input
L1	PoAC chain integrity	—	Structural	SHA-256 linkage, monotonic counter, timestamp freshness

Layer	Module	Code	Type	Signal
L2	`hid_xinput_oracle.py`	`0x28`	Hard cheat	HID report vs. XInput API discrepancy + gravity-absent signal (mean accel < 100 LSB)
L3	`tinyml_backend_cheat.py`	`0x29, 0x2A`	Hard cheat	9-feature temporal behavioral analysis (30→64→32→6 INT8 net)
L2B	`l2b_imu_press_correlation.py`	`0x31`	Advisory	IMU micro-disturbance absent in 5–80ms precursor window before button rising edge
L2C	`l2c_stick_imu_correlation.py`	`0x32`	Advisory	Max Pearson cross-corr of stick velocity vs. gyro_z at causal lags 10–60ms < 0.15
L4	`tinyml_biometric_fusion.py`	`0x30`	Advisory	11-signal Mahalanobis kinematic fingerprint: triggers, tremor FFT (8–12 Hz), touchpad biometric
L5	`temporal_rhythm_oracle.py`	`0x2B`	Advisory	CV < 0.08, Shannon entropy < 1.0 bits, 60 Hz quantization > 0.55; fires on $\geq 2/3$
L6	`l6_trigger_driver.py` + `l6_response_analyzer.py`		Advisory	Randomized trigger resistance challenge; human motor onset/settle/grip-variance curve

L2 — HID injection detection. Software injection attacks (SendInput, XInput emulation, vJoy, DS4Windows spoofing) cannot produce physical IMU readings. The oracle fires `0x28` when:

```
imu_noise = std(gyro) over 50 reports < 0.001 rad/s
AND
stick_magnitude > 0.15 of full range
```

The 0.001 rad/s threshold is below any real controller's noise floor (typically 0.01–0.05 rad/s at

rest). A physical controller always exceeds this floor due to hand micro-tremors. **Live measurement on DualShock Edge CFI-ZCP1:** stationary gyro std < 50 LSB ($\approx$ 0.05 rad/s, confirmed via `test_imu_noise_floor`); active play gyro std = 201 LSB ($\approx$ 0.22 rad/s) — a **10,000× margin** above the 0.02 LSB (0.001 rad/s) injection threshold.

Gravity-signal extension. A second, independent L2 signal fires on any session regardless of active-frame count: `mean(||accel||) < 100 LSB`. Real controllers under gravity always read $≈$2,048–2,150 LSB total accel magnitude; injected frames zero all three accel channels, producing magnitude $\approx$ 0. This signal closes the idle-start gap (sessions where the player was in a lobby with no active trigger inputs): validated at 100% injection detection after adding this signal (up from 80% pre-fix).

L3 — Behavioral ML. The 9-feature temporal classifier (velocity-stop events, jerk-correction lag, aim-settling variance, button timing σ^2, stick autocorrelation, reaction-time proxy) targets `MACRO` (σ^2 < 1.0 ms^2) and `AIMBOT` (ballistic jerk > 2.0) patterns that survive the L2 IMU check.

L4 — Biometric Mahalanobis fingerprinting. Eleven kinematic features per 50-report window are compared against a per-device *stable EMA baseline* — updated only on clean NOMINAL sessions to prevent fingerprint poisoning. The 11-feature space (Phase 17 expansion from 7): `trigger_resistance_change_rate`, `trigger_onset_velocity_L2/R2`, `micro_tremor_accel_variance`, `grip_asymmetry`, `stick_autocorr_lag1/5`, `tremor_peak_hz`, `tremor_band_power` (tremor FFT 8–12 Hz band), `touchpad_active_fraction`, `touch_position_variance`. The stable-vs-candidate architecture is the key security property: an adversary who gradually shifts the EMA over many borderline sessions cannot poison the stable reference.

`fingerprint_drift_velocity` (L2 norm between candidate and stable means) is an additional contamination signal surface-able via the `BridgeAgent` tool `get_behavioral_report`.

L5 — Temporal rhythm oracle. Bot scripts produce near-constant inter-press intervals. The oracle characterizes the inter-event timing distribution over a 120-event deque (min 20 samples) and fires `0x2B TEMPORAL_ANOMALY` when $\geq$ 2 of 3 signals are suspicious: - Coefficient of variation (CV) < 0.08 — near-zero timing variance - Shannon entropy < 1.0 bits — few distinct interval values (hardware-calibrated N=69, 3 players; human baseline: ~4.56 bits mean) - 60 Hz quantization score > 0.55 — intervals cluster at 16.67 ms multiples

L5 uses four-button tracking via `push_snapshot()` rising-edge detection: Cross (X), L2_dig, R2, and Triangle — in descending IBI-CV priority (1.373, 1.333, 1.176, 1.138). When no single button accumulates $\geq$ 20 inter-press intervals, a pooled IBI fallback merges all buttons with $\geq$ 5 samples into one sequence. This reduces genre-dependent exclusion from 24.6 % (R2-only, Phase 16B) to near-zero across N=69 real sessions. The `source` field ('cross', 'l2_dig', 'r2', 'triangle', 'pooled') is persisted in PITL metadata for operator visibility.

`rhythm_humanity_score = (cv_humanity + entropy_score + non_quant) / 3.0 □ [0,1]` contributes to the PHG humanity probability as a positive signal for high-variance, high-entropy timing.

L6 — Active physical challenge-response. L0–L5 are *passive* — they observe sensor signals generated by normal play. L6 is *active*: the host generates a randomized adaptive trigger resistance profile (one of 8 named profiles spanning Off, Rigid 80–255 N, Pulse slow/fast, asymmetric, and buildup patterns), writes it to the DualShock Edge via USB output report 0x02, and measures the controller's HID input stream for the human motor response over the following 3-second window.

Human motor response to a resistance change is governed by involuntary biomechanics: the hand requires 40–300 ms onset latency (neuromuscular delay + motor-planning time), produces measurable grip-force variance (accel magnitude variance > 0 from hand micro-tremors adjusting to the new resistance), and exhibits natural settling as muscles adapt. Software injection — which cannot feel resistance it cannot sense — cannot replicate any of these properties.

The classifier produces `p_human_L6 □ [0.0, 1.0]` from four metrics: - `onset_ms`: frames until trigger ADC delta > 5 LSB after challenge sent - `peak_delta`: max |r2_post − r2_pre_mean| in the response window - `settle_ms`: frames until r2 returns within 10% of pre-challenge mean - `grip_variance`: variance of `||accel||` during response window

Attack G (challenge-invariant injection) signature: `grip_variance == 0.0` (zeroed accelerometer) → `p_human_L6 = 0.0`; `onset_ms < 5 ms` (sub-neurological latency) → `p_human_L6 ≤ 0.2`. Both are impossible for physical human operation.

Safety invariants: L6 is disabled by default (`L6_CHALLENGES_ENABLED=false`). Challenges are never dispatched during idle windows (r2 = l2 = 0 for last 10 reports) to avoid disrupting gameplay. Triggers restore to BASELINE_OFF (no resistance) within 3 seconds and always on session shutdown. The null signal (no response received) returns `p_human_L6 = 0.5` — conservative, non-penalizing.

Current status: L6 is fully implemented and unit-tested (33 tests), including Attack G synthetic adversarial validation. Human motor response baseline calibration — characterizing real onset/settle/grip-variance distributions from real DualShock Edge challenge sessions — has not yet been performed. Section §10.6 describes this as the immediate next hardware validation milestone.

7.5.2.1 Game Genre Certification Requirements for PITL Biometric Activation

Each of the eleven L4 biometric features is conditional on specific controller usage patterns that vary substantially by game genre. A session in which the player never moves the right stick, never presses L2, and never triggers simultaneous dual-grip produces feature vectors with up to 9 of 11 fields structurally zero — not because the human is unusual, but because the game did not elicit the relevant motor behavior. This section derives minimum controller-usage requirements for each L4 feature, characterizes which game genres satisfy them, and defines VAPI tournament deployment certification tiers.

Empirical basis: Per-feature symmetric KL divergence computed from N=64 real DualShock Edge sessions (3 players, NCAA Football 26) in `docs/interperson-separation-analysis-v2.md §Phase 41`. In that dataset, 6 of 11 features are structurally zero and only `stick_autocorr_lag1/5` provides meaningful inter-player information — solely because Player 3 uses the right stick far less than Players 1/2, not because of individual physiological differences.

Minimum Controller-Usage Requirements per Feature **Table 7.5.2.1-A: L4 Feature Activation Requirements**

Feature	Required Controller Behavior	Minimum Threshold	Notes
trigger_resistance_change_count	Game must send mid-session adaptive trigger effect changes via USB output report 0x02	$\geq$ 1 effect-mode transition per session	Requires DualShock Edge firmware + game support. Structurally zero in any game with static trigger profiles.
trigger_onset_count_L2	L2 trigger must be depressed from rest (ADC $\leq$ 5) to engagement (ADC > 200) as distinct press events	$\geq$ 50 rising-edge events per session	NCAA Football 26: ~5–20 formation-select presses. Below threshold for onset statistics.
trigger_onset_count_R2	R2 trigger must exhibit onset cycles; continuous hold does NOT register new onset events	$\geq$ 50 rising-edge events per session	Sprint (hold) does not count. Requires repeated fire/brake/throw actions.

Feature	Required Controller Behavior	Minimum Threshold	Notes
`micro_tremor_accel`	Device must have still-frame windows (gyro_mag < 20 LSB) during active session; accel variance computed only during those frames	$\geq$ 10 still-frame passes per session	Present in most genres during brief pauses, menus, or low-movement moments. Gyro gate is empirically calibrated at 20 LSB (raw HID).
`grip_asymmetry`	L2 and R2 must both exceed 10 ADC simultaneously (dual-press frame)	$\geq$ 10 dual-press frames per session	Requires mechanical co-activation: ADS+shoot (FPS), brake+accelerate (racing), or parry+strike (action). Any game where L2 and R2 are contextually exclusive yields grip_asymmetry = 1.000 for all sessions.

Feature	Required Controller Behavior	Minimum Threshold	Notes
`stick_autocorr_lag1`	right_stick_x must deviate from dead-zone center (128) with temporal persistence	$\geq$ 100 non-dead-zone right-stick reports per session	Captures characteristic micro-correction patterns in sustained aim or camera movement. Spiky one-off movements contribute less than smooth persistent input.
`stick_autocorr_lag15`	Same as lag1, lag5 captures longer motor persistence	$\geq$ 200 non-dead-zone frames	Requires continuous right-stick engagement, not single-frame panning.

Feature	Required Controller Behavior	Minimum Threshold	Notes
`tremor_peak_hz`	right_stick_x must accumulate ≥ 513 non-dead-zone frames in the extractor's ring buffer	≥ 513 consecutive non-dead-zone reports (ring buffer fills in ~0.5s of continuous aim)	Physiological tremor is 8–12 Hz. Bot scripts that precisely track a target produce near-zero tremor. FFT resolution: 1.95 Hz/bin at 1000 Hz with 512 velocity samples.
`tremor_band_power`	Same as `tremor_peak_hz`	≥ 513 frames	Collapses to 0 when FFT is inactive. Correlated with `tremor_peak_hz`.

Feature	Required Controller Behavior	Minimum Threshold	Notes
`touchpad_active`	Session must be captured with Phase 17+ `capture_session.py` (adds `touch_active` field); player's resting thumb must contact touchpad	Post-Phase-17 capture, $\geq$ 1 touch-active frame	Field did not exist in pre-Phase-17 sessions. Touchpad contact is natural for thumb-resting posture on DualShock Edge; no explicit player action required.
`touch_position_variance`	Same as above; requires $\geq$ 3 touch-active frames for variance to be non-trivial	Post-Phase-17 capture, $\geq$ 3 touch-active frames	Captures per-player characteristic thumb resting position (high-value biometric once populated).

Game Genre Certification Table Certification tier is determined by the number of L4 features **active** in typical play for that genre. "Active" means the feature is expected to have non-zero variance across sessions from the same player — sufficient to inform the Mahalanobis fingerprint. Touchpad features (10, 11) are counted only for post-Phase-17 captures.

Tiers: - **FULL CERTIFICATION ($\geq$ 9/11):** L4 operates as a full inter-player biometric identifier. Intra-player anomaly detection and cross-player transplant detection are both reliable. - **STANDARD CERTIFICATION (6–8/11):** L4 inter-player discrimination is partial. Intra-player anomaly detection reliable; transplant detection viable with N $\geq$ 20 sessions per player. - **LIMITED CERTIFICATION (3–5/11):** L4 operates as intra-player anomaly detector only. Zero inter-player discriminability expected. L1, L2, L2B/C, L3, and L5 remain fully active for cheat detection; L4 adds session-consistency evidence

but cannot identify the player. - **NOT RECOMMENDED ($\leq$ 2/11):** L4 biometric layer is effectively inactive. PITL integrity is provided by L1/L2/L3/L5 only. Not suitable for tournament deployment where biometric identity binding is required.

Table 7.5.2.1-B: Per-Genre L4 Feature Activation

Genre	Rep. Titles	1 trg_resist	2 on-set_R2	3 on-set_L2	4 mi-cro_temp	5 temp_asym	6 au-to-corr	7 au-to-lag1	8 lag5	9 tremor	10 tpp_cv	11 tpp_var	**Active**	**Tier**
FPS / Battle Royale	COD Black Ops 6, Halo Infinite, Apex Legends, Fortnite	(!)[1]	✓	✓	✓	✓	✓	✓	✓	✓	✓[2]	✓[2]	**9–10**	**FULL**
Racing (Simulation)	Gran Turismo 7, Forza Motorsport, F1 24	✓[3]	✓	✓	✓	✓	(!)[4]	(!)[4]	(!)[4]	(!)[4]	✓[2]	✓[2]	**7–9**	**STANDARD–FULL**
Action-Adventure / RPG	Elden Ring, God of War, Spider-Man 2	(!)[5]	✓	✓	✓	✓	✓	✓	✓	✓	✓[2]	✓[2]	**8–10**	**FULL**

Genre	Rep. Titles	1 trg_resist	2 on-set_R2	3 on-set_L2	4 micro_trig	5 temp_asym	6 auto-lag	7 auto-lag	8 target_mod	9 tremor_mod	10 tip_force	11 force_var	Active	Tier
Adaptive Trigger Native	Ratchet & Clank: Rift Apart, Astro's Playroom	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓[2]	✓[2]	**10–11**	**FULL**
Fighting	Street Fighter 6, Mortal Kombat 1, Tekken 8	×	(!)[6]	(!)[6]	✓	×	×	×	×	×	✓[2]	✓[2]	**3–5**	(!) **LIMITED**
Sports — Real-Time Aim	NBA 2K25, MLB The Show 24 (pitching/hitting)	✓[3]	✓	✓	✓	✓	(!)	(!)	(!)	(!)	✓[2]	✓[2]	**6–8**	**STANDARD**
Sports — Sim Football (NCAA CFB, Madden)	NCAA Football 26, Madden NFL 25	×	×	×[7]	✓	×	×[8]	×[8]	×[8]	×[8]	✓[2]	✓[2]	**2–3**	(X) **NOT REC.** / (!) LIMITED

Genre	Rep. Titles	1 trg_res	2 onset_L2	3 onset_R2	4 micro_tremor	5 temporal_asym	6 autocorr_lag1	7 autocorr_lag5	8 tremor_amp	9 tremor_freq	10 touch_pp	11 touch_var	**Active**	**Tier**
Platformer / Narrative	Crash Bandicoot, Astro Bot, Death Stranding	×	(!)	✓	✓	×	×	×	×	×	✓[2]	✓[2]	**3–5**	(!) **LIMITED**

[1] FPS games vary: COD on PlayStation uses adaptive trigger support (Haptic Feedback mode); PC-origin ports (Apex, Fortnite) typically do not. [2] Touchpad features require post-Phase-17 session capture. For pre-Phase-17 captures, subtract 2 from active count. [3] Gran Turismo 7, Forza Motorsport, and NBA 2K use resistance profiles for ABS/grip feedback; feature activates reliably. [4] Racing right-stick (camera) is often held static during race focus. Feature activates in camera-heavy moments (replays, cornering). Autocorr and tremor are partial. [5] Game-dependent; adaptive trigger support varies. God of War: no. Horizon FW: yes (bow draw). Spider-Man 2: yes (web-shooter modulation). [6] L2/R2 are medium/heavy attacks in many fighting games; $≥$50 presses per session is achievable in high-level play but not guaranteed at lower engagement. [7] R2 is sprint in NCAA Football 26 and is held continuously, not pressed in repeated onset cycles. Onset velocity requires release-and-re-press events. [8] Right stick is near-stationary (128 dead zone) throughout almost all NCAA Football 26 sessions. From N=64 sessions: only 1/38 P1 sessions had any non-zero `tremor_peak_hz`.

NCAA Football 26: LIMITED CERTIFICATION Based on N=64 hardware-validated sessions (Phase 41 post-analysis), NCAA Football 26 with the DualShock Edge is assigned **LIMITED CERTIFICATION** for VAPI tournament deployment.

What the protocol proves in this configuration: - L1 PoAC chain integrity — every record cryptographically signed, ordered, and linked. Tampering, reordering, and omission are detectable. - L2 injection detection (`0x28`) — IMU noise floor and gravity-absent signal fully active. Software injection is detected regardless of game. - L2B/L2C advisory oracles (`0x31`, `0x32`) — IMU-button causal coupling and stick-IMU temporal cross-correlation. Functional for button events; L2C is limited by right-stick dead-zone (68/69 sessions static stick, both confirmed in N=69 Phase 17 and N=64 Phase 41 data). - L3 behavioral ML (`0x29`, `0x2A`) — Fully active. Bot macro/aimbot patterns are detectable. - L5 temporal rhythm oracle (`0x2B`) — Fully active on Cross, L2_dig, R2, Triangle button inter-press intervals. Multi-button pooled fallback covers low-press-frequency play styles.

What L4 provides in this configuration: L4 operates as **intra-player session consistency detector only** — not as an inter-player identifier.

Property	Status in NCAA CFB 26
Intra-player session anomaly detection (is this session consistent with this device's history?)	**ACTIVE** — 3 active features (micro_tremor, stick_autocorr_lag1/5) inform the EMA fingerprint. Unusual play behavior within a single player's session history is detectable.
Cross-player transplant detection (is this a different player using this credential?)	**NOT ACTIVE** — separation ratio 0.362 (threshold > 2.0). P1/P2 are statistically indistinguishable across all 5 active features. L4 cannot detect one player's sessions submitted under another's device ID.
Structural zero-variance features (trigger_resistance_change_rate, trigger_onset_velocity_L2/R2, tremor_peak_hz, tremor_band_power)	**INERT** — auto-excluded by ZERO_VAR_THRESHOLD = 1e-4 in BiometricFusionClassifier. These features do not contribute false-positive 0x30 signals; they simply do not contribute.
Touchpad features (touchpad_active_fraction, touch_position_variance)	**PENDING** — zero in all pre-Phase-17 sessions. Will populate in next capture with Phase 17+ script. Expected to add one stable per-player discriminator once populated.

Recommended use: VAPI-certified NCAA Football 26 tournaments are appropriate where the integrity requirements are: (a) verifying that a real controller was physically operated (not software-injected), (b) detecting automated bot scripts via L5 timing anomalies, and (c) establishing a tamper-evident PoAC evidence chain per session. They are **not** appropriate where the requirement is to verify that the same human operated the device across all sessions — that guarantee requires FULL or STANDARD CERTIFICATION (FPS, Racing, Action-Adventure genres).

Upgrade Path to FULL CERTIFICATION for Football Titles Any future football title that implements the following mechanics will upgrade from LIMITED to STANDARD or FULL CERTIFICATION without protocol changes:

1. **Implement adaptive trigger resistance profiles** for kick-power meters, tackle strength, or catch trajectories. This activates `trigger_resistance_change_rate`.
2. **Require L2 as a non-formation action** (e.g., receiver route adjustments post-snap, secondary defensive commands). This produces $≥$50 L2 onset events per session and activates `trigger_onset_velocity_L2`.
3. **Map right stick to a continuous action** (spin move, juke direction, camera rotation during replays with player control). This fills the 513-frame FFT ring buffer and activates `tremor_peak_hz`, `tremor_band_power`, and strengthens `stick_autocorr_lag1/5`.
4. **Enable simultaneous L2+R2 mechanics** (e.g., a precision kick charge requiring both triggers). This activates `grip_asymmetry`.

Alternatively, a VAPI-certified multi-genre session structure (one FPS session + one CFB session per hour) would satisfy the feature activation requirements through the FPS session, with the CFB session contributing L1/L2/L3/L5 evidence to the same device's PHG credential stream.

Derived from per-feature KL divergence analysis in `docs/interperson-separation-analysis-v2.md §Phase 41 Post-Analysis`. Separation thresholds based on Gaussian KL divergence $\geq$ 0.5 = discriminating, Cohen's d $\geq$ 0.5 = meaningfully separated. Game genre assignments are protocol-author assessments based on published control schemes; individual game configuration may vary.

7.5.3 Zero-Knowledge PITL Session Proof

The bridge generates a Groth16 proof (BN254, ~1,820 constraints, 2^11 powers-of-tau) at session shutdown, establishing four cryptographic invariants without revealing raw sensor features on-chain:

Public inputs (5): - `featureCommitment` — Poseidon(7)(scaledFeatures[0..6]) - `humanityProbInt` — humanity_prob × 1000 ∈ [0, 1000] - `inferenceResult` — 8-bit inference code (see known limitation below) - `nullifierHash` — Poseidon(deviceIdHash, epoch) — anti-replay binding - `epoch` — block.number / EPOCH_BLOCKS

Circuit constraints enforced: - C1: featureCommitment = Poseidon of exactly the 7 secret L4 biometric features - C3: humanityProbInt ∈ [0, 1000] - C4: nullifierHash = Poseidon(deviceIdHash, epoch) — session uniqueness - C5: L5 rhythm score ∈ [0, 1000]

Known limitation — inference code not bound on-chain: Circuit constraint C2 (inferenceResult range check) exists in the circom circuit, but `PITLSessionRegistry.sol` passes `pub[2] = 0` to the on-chain verifier. The inference code is committed off-chain in the bridge SQLite store (`records.inference`). This means the ZK proof does not currently prevent the bridge from submitting a proof with any inference code — the circuit cannot distinguish. This is documented in `PITLSessionRegistry.sol` and will be corrected when the circuit is upgraded.

What the proof does NOT establish: - That the raw sensor data was transformed into features correctly - That the bridge executed the biometric pipeline honestly end-to-end - That the inference code in the PoAC record matches the inference the circuit was given

`PITLSessionRegistry.sol` (deployed at `0x8da0A497234C57914a46279A8F938C07D3Eb5f12`, testnet) accepts 256-byte proofs, enforces `usedNullifiers` anti-replay, and tracks per-device `latestHumanityProb` and `sessionCount`. The real Groth16 verifier (`PitlSessionProofVerifier`, `0x07D3ca1548678410edC505406f022399920d4072`) is set and active; mock mode is disabled on the live deployment.

The trusted setup used the `run-ceremony.js` single-contributor development ceremony. A production deployment requires a multi-party MPC ceremony (§10.3).

7.5.4 PHG Credential and Economic Enforcement

PHG humanity probability fusion. Per session, three signals are fused into `humanity_probability` □ `[0,1]`:

```
p_L4   = exp(-max(0, d_L4 - 2.0))            # biometric match (11-signal)
p_L5   = rhythm_humanity_score                    # timing humanity
p_E4   = exp(-drift / 3.0)                   # cognitive stability
p_L2B    =  imu_press_oracle.humanity_score()             #  IMU-button causal coupling [0,1]
```

```
p_L2C  = stick_imu_oracle.humanity_score()      # stick-IMU cross-
correlation [0,1]

# Without L6 (default):
humanity_probability = 0.28·p_L4 + 0.27·p_L5 + 0.20·p_E4 + 0.15·p_L2B + 0.10·p
```

L6 reweighting. When L6 is active, the humanity probability fuses six independent signals: `p_human = 0.23·p_L4 + 0.22·p_L5 + 0.15·p_E4 + 0.15·p_L6 + 0.15·p_L2B + 0.10·p_L2C`. When L6 is disabled (default), the five-signal formula above applies. L2B and L2C default to 0.5 (neutral) before oracle warmup, preserving [0,1] boundedness. L6 participation in the ZK PITL circuit (§7.5.3) is noted as future work pending the multi-contributor ceremony (§10.3).

PHG score weighting. PHG score deltas are weighted by `humanity_probability` (+50% bonus at p=1.0) and by behavioral analysis: `delta × max(0.0, 1.0 − warmup×0.8 − burst×0.5)`. This makes the on-chain PHG score reflect *quality* of human activity, not merely volume.

On-chain credential lifecycle. `PHGCredential.sol` (soulbound, ERC-5192-inspired): - Minted: `mintCredential(deviceId, nullifierHash, featureCommitment, humanityProbInt)` when a device has both a PHG checkpoint and a PITL session proof - Active: credential contributes to `TournamentGateV3.isEligible()` - Suspended: `suspend(deviceId, evidenceHash, durationSeconds)` — provisional revocation by bridge when `InsightSynthesizer` Mode 5 fires - Reinstated: `reinstate(deviceId)` — automatic when device is labeled `cleared`

PHGCredential suspension (Mode 5 enforcement). `InsightSynthesizer._synthesize_credentia` runs each 6-hour cycle: - For each device labeled `critical`: increment `consecutive_critical` counter - If `consecutive_critical ≥ 2` AND has active credential AND not already suspended: compute suspension duration = `min(base_s × 2^(consecutive-2), max_s)` where `base_s = 604800` (7d), `max_s = 2419200` (28d) - Call `chain.suspend_phg_credential(device_id, evidence_hash, duration_s)` - Evidence hash = `SHA-256(f"{device_id}:{digest_id}")` — anchored to immutable digest row - For each device labeled `cleared`: call `chain.reinstate_phg_credential()` - For each device labeled `stable` or `warming`: reset `consecutive_critical` to 0

Tournament gate hierarchy: - `TournamentGateV1` — minimum PHG cumulative score only - `TournamentGateV2` — cumulative score AND recent velocity (time-qualified) - `TournamentGateV3` — V2 gates AND `PHGCredential.isActive()` (suspension-aware)

Biometric-anchored session continuity. A player who gets a new controller inherits their PHG history if: `diagonal_mahalanobis(old_fingerprint, new_fingerprint) < 5.369` (the continuity threshold is tighter than the 7.019 anomaly threshold; both derived from the N=69 calibration corpus). The PHG score is transferred on-chain via `PHGRegistry.inheritScore()`; the source is zeroed to prevent double-counting. Each device can be a continuity source and destination exactly once.

7.5.5 Longitudinal Intelligence and Adaptive Feedback

Behavioral archaeology. `BehavioralArchaeologist` performs longitudinal analysis of per-session PITL readings using pure numpy (no LLM, always available):

Warm-up attack detection:

```
warmup_score = σ(drift_slope × humanity_slope × 20000 − 1.0)
```

where σ is sigmoid and slopes are computed by least-squares regression over record index (invariant to adversarial timestamp manipulation). Stable human players: warmup_score < 0.3. Coordinated warm-up (rising drift + rising humanity over sessions): warmup_score > 0.7.

Burst farming detection:

```
burst_score = min(1.0, CV_inter_checkpoint / 2.0)
```

High gap-CV indicates bursty PHG checkpoint accumulation followed by inactivity.

Biometric certificates (long-term): - `biometric_stability_cert`: mean drift velocity < 0.5 over last 20 sessions - `l4_consistency_cert`: L4 distance CV < 0.3 over last 20 sessions

Network correlation detection. `NetworkCorrelationDetector` builds an N×N pairwise biometric distance matrix across all fingerprinted devices, then applies BFS DBSCAN (ε = 1.0, min_samples = 3). Devices in clusters where `avg_intra_distance < ε/2` are flagged as potential bot farms: genuine human players have idiosyncratic kinematic profiles; bots running the same automation software cluster tightly in biometric space.

```
farm_suspicion_score = min(1.0, (size−2)/5 + (ε−avg_d)/ε)
```

InsightSynthesizer (6-hour cycle). Six synthesis modes run independently each cycle: - **Mode 1**: Rolling 24h/7d/30d window digests (bot_farm_count, high_risk_count, federated_count, anomaly_count, dominant_severity, top_5_devices, narrative text) - **Mode 2**: Per-device risk trajectory labels via deterministic state machine: `_risk_label(bot, high_risk, fed, anomaly, prior) → {stable, warming, critical, cleared}` - **Mode 3**: Federation topology fingerprints for clusters confirmed across $≥$2 bridge instances - **Mode 4**: Detection policy synthesis — translates risk labels into L4 threshold multipliers: `{critical: 0.70, warming: 0.85, stable: 1.00, cleared: 1.00}` - **Mode 5**: PHGCredential enforcement (suspension / reinstatement) as described in §7.5.4 - **Mode 6**: Living calibration — exponential decay weighted threshold evolution from NOMINAL records (Phase 38)

Adaptive L4 feedback loop. After `classify()` returns None (no hard cheat), the bridge checks the device's active detection policy:

```
if multiplier < 1.0:
    effective_thresh = 3.0 * multiplier
    if last_distance > effective_thresh:
        synthesize(0x30, confidence=min(255, 180 + int(excess*30)))
```

The loop is bounded (floor multiplier = 0.5 → max tightening 50%), reversible (label change on next synthesis cycle auto-restores threshold), and non-fatal (policy lookup always wrapped in bare `except Exception: pass`). Hard cheat codes `0x28/0x29/0x2A` are never affected by this mechanism.

Mode 6 — Living Calibration (Phase 38). Every 6-hour synthesis cycle, `InsightSynthesizer._synthesize_living_calibration()` recomputes L4 Mahalanobis thresholds from the last 200 NOMINAL warmed records using exponential decay weighting (α=0.95, index 0 = most recent = weight 1.0):

```
weights[i] = 0.95^i / sum(0.95^j for j in range(n))
w_mean = sum(d[i] * weights[i])
w_std  = sqrt(sum(weights[i] * (d[i] - w_mean)^2))
candidate_anomaly    = w_mean + 3.0 * w_std
candidate_continuity = w_mean + 2.0 * w_std
```

Updates are bounded to ±15% per cycle to prevent oscillation, with a minimum threshold floor of 3.0. Updates are applied live to the running config without a bridge restart.

Per-player profiles are computed for devices with $≥$30 NOMINAL records: - `personal_anomaly = min(w_mean + 3.0 * w_std, global_anomaly)` — tighter-than-global, enforced by `min()` - `personal_continuity = min(w_mean + 2.0 * w_std, global_continuity)` - Profiles persist in the `player_calibration_profiles` SQLite table

During L4 classification, `DualShockIntegration._get_effective_l4_threshold(device_id)` returns `min(global_threshold, personal_threshold)`, fetching personal profiles from a 6-hour cache. Per-player thresholds can only tighten detection — they can never loosen it.

Calibration health is self-monitored every cycle: (1) stale data alert if newest NOMINAL record is >48h old; (2) distribution shift alert if recent 20 records differ from historical 80 by >25% of mean. Both fire as `calibration_health_*` insights in the `protocol_insights` table.

7.5.6 Federated Cross-Instance Threat Correlation

Bot farms that distribute devices across multiple bridge shards stay below each instance's local detection threshold. `FederationBus` counters this via privacy-preserving cluster fingerprint exchange.

Privacy model:

```
cluster_hash = SHA-256("|".join(sorted(device_ids)))[:16]
bridge_id = SHA-256(f"bridge:{api_key}")[:16]
```

Raw device IDs never leave the originating bridge. The `GET /federation/clusters` endpoint returns only `is_local=True` records, preventing echo amplification in hub-and-spoke topologies.

Cross-confirmation:

```sql
SELECT cluster_hash, COUNT(DISTINCT bridge_id) AS peer_count
FROM federation_registry
GROUP BY cluster_hash
HAVING peer_count >= 2
```

A single misbehaving bridge cannot inflate the count by using `COUNT(DISTINCT bridge_id)` rather than `COUNT(peer_url)`.

On-chain anchor: `FederatedThreatRegistry.sol` stores cross-confirmed hashes immutably. `MultiVenueConfirmed` is emitted at `_reportCount ≥ 2`; anti-replay via `_hasReported[clusterHash][reporter]`.

7.5.7 BridgeAgent and Alert Dispatch

`BridgeAgent` (`claude-sonnet-4-6`) exposes natural-language operator intelligence through 17 deterministic tool bindings over bridge data, a Server-Sent Events streaming endpoint (`GET /operator/agent/stream`), and an autonomous `react()` path that interprets `BIOMETRIC_ANOMALY` and `TEMPORAL_ANOMALY` events without operator input. All high-frequency detection (L2–L5) remains deterministic; the LLM operates only at the human-paced query layer where synthesis latency is acceptable. Session history persists across restarts in SQLite. `AlertRouter` complements the agent by polling `protocol_insights` every 30 seconds and dispatching events meeting the configured severity threshold to an operator webhook (Slack, PagerDuty, or generic JSON) via stdlib `urllib`, with no new dependencies and non-fatal failure handling. See Appendix B for the complete tool catalogue and streaming interface specification.

7.5.9 DePIN Extensibility Validation

The IoTeX Pebble Tracker (nRF9160 SiP, ARM Cortex-M33 @ 64 MHz, CryptoCell-310) validates protocol extensibility. The same 228-byte PoAC wire format, the same three-layer agent architecture, and the same on-chain contract stack operate unchanged. Only the sensor commitment schema differs (schema v1, environmental: BME680 temperature/VOC, ICM-42605 IMU, TSL2572 lux, GPS) versus the DualShock Edge (schema v2, kinematic/haptic).

This confirms VAPI's core design claim: the verification mechanism is device-agnostic; the detection surface is device-specific.

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8. Evaluation

8.1 Cryptographic Overhead

Table 4: Per-Operation Latency on CryptoCell-310 (nRF9160)

Operation	Latency
SHA-256 (96 B sensor buffer)	0.8 ms
SHA-256 (1.5 KB world model)	2.1 ms
SHA-256 (model manifest)	1.4 ms
ECDSA-P256 sign	6.2 ms
NVS write (36 B)	1.8 ms
Total PoAC generation	**12.3 ms**

These figures are estimated via cycle-accurate emulation. Real-hardware validation on a physical nRF9160 DK is future work. The 12.3 ms total overhead is 0.041% of the DePIN 30-second reflexive cycle budget.

8.2 On-Chain Gas Costs

Operation	Gas per record
`verifyPoAC` (individual)	148,230
`verifyPoACBatch` (batch of 10)	81,245
`PHGRegistry.commitCheckpoint`	~72,000
`PHGCredential.mintCredential`	~110,000
`PHGCredential.suspend`	~85,000
`FederatedThreatRegistry.reportCluster`	~65,000

At IoTeX's typical gas price of 1 Gwei and IOTX at $0.03, a batch of 10 verifications costs approximately $0.00024. The P256 precompile at `0x0100` is essential: without it, P256 verification in pure Solidity requires ~1.2M gas per signature.

8.3 Anti-Cheat Detection — Real Hardware Adversarial Validation

Detection benchmarks were produced by running six deterministic attack transforms against 55 real 1000 Hz DualShock Edge sessions (N=50 human baseline, 2026-03-02). Each adversarial session is a reproducible transformation of a real captured session targeting the specific physical property each PITL layer relies on.

Table 5: Adversarial Detection Matrix — Real Hardware (DualShock Edge CFI-ZCP1, 1000 Hz)

Attack Type	N	L2 Det%	L4 Det%	L5 Det%	Any%	Notes
Replay (chain-level)	5	0%	20%	0%	20%	Chain attack — PITL not the right layer; on-chain replay prevention handles this
IMU-stripped injection	10	**100%**	0%	0%	**100%**	L2 gravity signal + active-frame gyro; zeroed accel always detectable

Attack Type	N	L2 Det%	L4 Det%	L5 Det%	Any%	Notes
Perfect-timing macro	10	50%	30%	**100%**	**100%**	L5 CV=0.0054, entropy=0.22 bits — far below human baseline
Biometric transplant	5	0%	0%	0%	0%	Single-person dataset: chimeric fingerprint within one person's Mahalanobis ball
Gradual warmup (bot→human)	10	20%	0%	60%	60%	Sessions 1–6 ($\alpha \leq 0.56$) detected; sessions 7–10 reach near-human territory
Quantization-masked bot	15	67%	0%	**100%**	**100%**	Mean quant_score=0.898; entropy collapses to ~0 bits despite timing jitter

Table 6: L5 Human Baseline — Hardware Calibrated (N=69 Sessions, 3 Players, DualShock Edge)

Metric	Human Mean	Human 10th Pct	Threshold	Safety Margin
CV (std/mean)	1.184	0.789	< 0.08	**9.9× vs 10th pct**

Metric	Human Mean	Human 10th Pct	Threshold	Safety Margin
Shannon entropy	2.085 bits	1.231 bits	< 1.0 bits	**1.2× vs 10th pct**
Quant score	0.590	0.423	> 0.55	— (2/3 rule prevents FP)

Human quant score (mean 0.59) slightly exceeds the threshold because humans also time button presses to game-loop frame boundaries. The 2/3-signal requirement prevents false positives since CV and entropy remain far on the human side.

Table 7: L4 Biometric — N=69 Production Thresholds (11-feature space, Phase 17)

Scenario	Mahalanobis d	L4 Fires?
Same human, different session (hw_* baseline, N=69, 3 players)	mean 2.07, max ~7.0	No (threshold **7.019**)
Genuine biometric outlier (2/69 sessions)	> 7.019	Yes — expected at 3σ
Bot farm (transplant, same person)	Within personal ball	No — requires multi-person dataset

Human false positive rate: 2.9% (2/69 sessions). Two sessions exceed the mean+3σ threshold — expected at the 3σ level. Threshold derived from 69 real sessions across 3 distinct players (dist_mean=2.068, dist_std=1.650; threshold = mean+3σ = 7.019). N=69 is the full calibration corpus spanning Players 1–3 (hw_005–hw_073).

Stationary control baseline. A 30-second session with the controller untouched on a desk (sessions/adversarial/stationary_control_001.json, 999.7 Hz) confirms: - PITL result: NOMINAL - Gyro std at rest: 1.3–1.5 LSB; P95 gyro magnitude: 9.54 LSB - Mean accel magnitude: ~2150 LSB (gravity); **14,000× injection detection margin**

Known limitation. The biometric transplant attack (0% detection in the N=50 single-player adversarial suite) requires a multi-person calibration dataset to overcome. The updated N=69 calibration corpus now spans 3 distinct players (hw_005–hw_073); inter-person Mahalanobis separation has been computed (separation ratio 0.362 — see §8.6 and §10.7). The result is honest: L4 does not currently separate players with the current feature set and calibration corpus.

L6 Active Challenge-Response. The L6 layer is implemented and unit-tested (§7.5.2). Live adversarial hardware benchmarking of L6 — running randomized challenge profiles against a physically connected DualShock Edge with real human play and software-injection replay — has not yet been performed. Human response baseline statistics (onset latency distribution, grip-variance distribution) are required before L6 detection thresholds can be empirically calibrated. This is the immediate next hardware validation task (§10.6).

Attack-to-layer mapping. Not every adversarial category is handled by PITL layers L2–L6. The adversarial benchmark results must be read with the correct layer assignment:

- **Replay (20% PITL hit) — chain-level attack handled by L1/PoAC, not PITL.** Replay is correctly detected by the PoAC monotonic counter and hash-linkage integrity check (L1). The 20% "detection" recorded in L4 during replay tests is statistical noise from biometric features slightly deviating from the enrolled baseline when frames are reordered — not a reliable signal. PITL layers L2–L6 are not designed for replay detection and should not be cited as replay mitigations.
- **Warmup (60%) — fundamental limitation of any gradual-onset attack.** Sessions 7–10 of the warmup attack are constructed to interpolate toward human behavior. A sufficiently patient attacker who degrades metrics slowly enough will eventually produce a session indistinguishable from legitimate skill improvement. BehavioralArchaeologist catches early-to-mid progression (sessions 1–6 reliably), but the attacker "graduates" as the session window saturates with near-human samples. This is a known limitation of any session-scoped behavioral detector. The explicit defense-in-depth is Mode 5 credential suspension (§7.5.4), which operates across sessions rather than within a single session, flagging progressive drift over time.
- **Transplant (0%) — architectural constraint, not a detection failure.** With a single player's calibration data, every session falls within the same Mahalanobis ball by construction. Transplant detection requires multi-person calibration: a second player with 5–10 sessions would allow the system to demonstrate either inter-person separation (validating the biometric fingerprint) or reveal that the feature set needs augmentation. The inter-person separation result for the N=69 3-player corpus (ratio 0.362) is documented in §10.7 rather than as a detection failure.

8.4 DualShock Edge Hardware Validation

28 hardware tests run on a physical Sony DualSense Edge (CFI-ZCP1) in five suites provide real-device empirical data for the key PITL detection surfaces. The test session used a CFI-ZCP1 connected via USB-C to a Windows 11 machine; `hidapi` enumerated the device at VID=0x054C, PID=0x0DF2, interface 3 (usage_page=1, usage=5).

Table 8: Live Hardware Measurements — DualShock Edge CFI-ZCP1 (USB mode)

Measurement	Value	Spec / Expected	Test
USB polling rate	**1002 Hz**	1000 Hz ± 15%	`test_1_polling_rate`
Accel magnitude (stationary, 1g ref)	**8267 LSB**	~8192 LSB (1g)	`test_7_micro_tremor`
Accel magnitude variance (held, natural grip)	**278,239 LSB²**	> 0 (injects: 0)	`test_7_micro_tremor`
Gyro noise std (active play)	**201.65 LSB**	> 0.02 LSB threshold	`test_5_imu_stick_co`
Gyro noise std (stationary)	**< 50 LSB**	< 50 LSB (pass)	`test_imu_noise_floo`
Injection detection margin	**14,000×** above stationary-control baseline	—	Derived from stationary_control_001.json
Report counter violations	**0 / 200 reports**	0	`test_2_report_count`
Sensor commitment v2 (SHA-256)	**Deterministic**	Required	`test_5_sensor_commi`
Distinct commitments (distinct reports)	**4 / 4**	All distinct	`test_5_sensor_commi`

Measurement	Value	Spec / Expected	Test
Timestamp field (bytes 12–14)	**49/49 advancing**	> 80%	`test_4_timestamp_fi`

The **14,000× injection detection margin** is derived from the stationary control baseline (sessions/adversarial/stationary_control_001.json): mean accel magnitude ~2150 LSB (gravity present) versus software injection which zeros all three accel channels (magnitude ≈ 0). The gyro dimension provides a separate **10,000× margin**: software injection produces gyro std ≈ 0 LSB; the physical device at rest produces < 50 LSB and in active play > 200 LSB. Any threshold between 0.02 LSB and 50 LSB provides reliable separation with zero false positives on real hardware.

The **278,239 LSB^2 accel variance** measured during normal hand-held use (no deliberate motion) demonstrates that the micro-tremor signal exists at meaningful amplitude in natural play conditions — not merely during controlled vibration or aggressive movement. This validates micro-tremor as a practical biometric feature for everyday detection, not a laboratory artifact.

Report-counter monotonicity confirms 200 consecutive reports with zero gaps or violations on Windows 11 USBHID. The polling rate of 1002 Hz confirms 1 kHz resolution for the 50-report feature extraction window assumed throughout the L4/L5 pipeline.

Test suite structure (28 tests):

File	Tests	Type
`test_dualshock_live.py`	6	HID enumeration, format, axes, IMU floor, commitment
`test_pitl_live.py`	5	PITL transport smoke (report volume, chain, features)
`test_dualshock_report_timing.py`	5	1 kHz rate, counter, gaps, timestamp, wrap
`test_dualshock_biometric.py`	7	L4 fusion, stable-track quarantine, micro-tremor, IMU-stick coupling, trigger onset velocity
`test_dualshock_adaptive_triggers.py`	5	Trigger ADC range, effect byte readback, release return, independence, sensor_commitment_v2

All tests include embedded step-by-step physical procedures (timing guidance, action prompts) so that operators without code expertise can execute the full hardware validation protocol. Tests are gated behind `@pytest.mark.hardware` and excluded from CI by default.

N=69 Biometric Calibration (2026-03-07, 3 Players). Following the 28-test hardware validation, 69 sessions across 3 distinct players were captured and used to calibrate all PITL thresholds empirically. The calibration corpus spans: - Player 1 (self): hw_005–hw_044 (38 sessions) - Player 2: hw_045–hw_058 (14 sessions) - Player 3: hw_059–hw_073 (12 sessions; 5 excluded for anomalous polling)

Phase 17 extended the L4 feature space from 7 to 11 features (adding tremor FFT 8–12 Hz band power/peak and touchpad biometrics). Three of the eleven features are structurally zero across all N=69 sessions (trigger_resistance_change_rate, touchpad_active_fraction, touch_position_variance) and are auto-excluded from calibration.

Threshold	Design-time estimate	Hardware-calibrated (N=69, 3 players)
L4 anomaly (ANOMALY_THRESHOLD)	3.0	**7.019** (mean+3σ, 11-feature; 7-feature: 6.905)
L4 continuity (CONTINUITY_THRESHOLD)	2.0	**5.369** (mean+2σ, 11-feature; 7-feature: 5.190)
L5 entropy	1.5 bits	**1.0 bits** (human 10th pct: 1.231 bits)
L5 CV	0.08	0.08 (unchanged; human mean: 1.184, 10th pct: 0.789)
L2B coupled_fraction	—	**0.55** (human mean: 0.786; 64/69 sessions with $≥$15 presses)
L2C max_causal_corr	—	**0.15** fixed threshold (0/69 false positives after abs() fix)

Calibration confidence: **HIGH** (N=69, 3 players). Values encoded as defaults in `controller/tinyml_biometri` and overridable via environment variables. See `calibration_profile.json` for the full calibration record.

The adversarial validation suite (`scripts/run_adversarial_validation.py`) subsequently validated these thresholds against 55 adversarial sessions across 6 attack types — see §8.3 for the full detection matrix.

8.5 Test Coverage Summary

Suite	Count	Scope
Bridge pytest	865	Full pipeline (asyncio bridge, store, agent, enforcement, federation, L6, L2B/L2C/CalibAgent, living calibration Phase 38, multi-button L5 Phase 39)
SDK pytest	28	Self-verifying client SDK (chain hash fix: verify_chain_link now uses SHA-256(164B body))
Hardhat	354	All Solidity contracts
Hardware	28	Physical DualShock Edge (gated `@pytest.mark.hardware`, excluded from CI)
E2E	14	End-to-end simulation (requires Hardhat node; excluded from CI)

Suite	Count	Scope
Total	**~1,289**	*~1,261 in CI (excluding 28 hardware, 14 E2E counted separately)*

Note: Phase 17 added 45 new bridge tests: 18 for `l2b_imu_press_correlation` (L2B IMU-button causal latency oracle), 15 for `l2c_stick_imu_correlation` (L2C stick-IMU cross-correlation oracle), and 12 for `calibration_agent` (auto-calibration threshold agent). The +33 L6 tests cover `l6_challenge_profiles`, `l6_trigger_driver`, `l6_response_analyzer`, and L6 integration including Attack G adversarial detection. Phase 38 added tests for Mode 6 living calibration and the `get_calibration_status` BridgeAgent tool.

8.6 What VAPI Does Not Yet Validate

The following detection capabilities are implemented but lack empirical calibration, or are explicitly not yet implemented:

L6 human response baseline. L6 Active Physical Challenge-Response is fully implemented (33 tests, 8 trigger profiles). Threshold parameters (onset_threshold_ms, settle_threshold_ms, classification weights) are derived from biomechanical priors, not measured distributions on real DualShock Edge players. False positive and false negative rates are unknown. L6 is disabled by default (`L6_CHALLENGES_ENABLED=false`) and must not be used as a primary tournament gate until N$≥$50 real challenge sessions are collected and analyzed.

Inter-person biometric identification. L4 is an intra-player anomaly detector (separation ratio 0.362, below the 1.0 threshold required for reliable identification). It detects deviation from a player's own baseline; it does not identify *who* the player is. This is correct positioning for the current feature set: three of eleven features are structurally zero across all N=69 calibration sessions (trigger_resistance_change_rate, touchpad_active_fraction, touch_position_variance). The honest interpretation: L4 catches sessions that are anomalous for *this device's history*, not sessions that belong to *a different person*.

ZK inference code binding. The Groth16 circuit has a constraint on inferenceResult range, but the on-chain verifier receives pub[2]=0, making this constraint trivially satisfied. The inference code in PoAC records is committed off-chain only.

Professional bot software. No commercial aimbot software, ML-driven bot inputs, or game-specific macro tools have been used as labeled adversarial data. All adversarial benchmarks use deterministic transforms of real human sessions.

Bluetooth transport calibration. BT transport is implemented (transport-aware parsing, L0 presence verifier, separate config thresholds) but all N=69 calibration sessions were captured via USB. L4/L5 thresholds carry no empirical grounding for Bluetooth polling rates (125–250 Hz).

Bridge as trusted intermediary. Despite ZK constraints on the biometric pipeline, the bridge remains operationally trusted: it controls which records are submitted on-chain, can withhold records (detectable via chain gaps), and computes humanity scores without end-to-end ZK coverage of the raw→feature transformation. The ZK proof constrains computation *given features*; it does not verify that the features were computed correctly from raw sensor data.

9. Security and Threat Model Analysis

9.1 Threat Mitigations

T1 — Record fabrication. ECDSA-P256 private key resides in CryptoCell-310 PSA persistent secure storage (key ID `0x00010001`), accessible only from the Secure partition. The on-chain verifier checks signatures via the P256 precompile.

T2 — Replay attacks. Monotonic counter (persisted in NVS flash) strictly increases across power cycles. `PoACVerifier` enforces `ctr > chainState.lastCounter`. ZK PITL proofs use `nullifierHash = Poseidon(deviceIdHash, epoch)` — binding proofs to one device × one epoch.

T3 — Selective omission. Hash-chain linkage makes omission detectable: a gap in counter values signals omission even without the hash link. The bridge can withhold records but cannot produce an alternative chain without the signing key.

T4 — Synthetic sensor injection. PoAC constrains but does not fully prevent this attack. The adaptive trigger resistance state byte (read from controller ADC, not writable from host HID stack, §7.5.1) narrows the viable injection surface. The L4 biometric stable-track fingerprint and the L5 timing distribution provide behavioral constraints: an adversary must maintain consistent synthetic input across a 32-observation world model, produce plausible classification outputs, and sustain deception across the L4 stable-track update window without triggering drift velocity alerts.

L6 additional mitigation. Active physical challenge-response (§7.5.2 L6) adds an orthogonal attack surface: software injection cannot produce human-biomechanical grip adjustment responses (onset latency 40–300 ms, accel variance > 0) in reaction to resistance profiles it cannot sense. An injector that also observes and attempts to synthesize L6 responses must simultaneously produce plausible IMU variance (L2), consistent biometric kinematics (L4), and non-degenerate timing distributions (L5) — a compound constraint that dramatically narrows the viable attack surface.

T5 — Fraudulent bounty claims. `submitEvidence()` requires a PoAC record hash in `PoACVerifier.verifiedRecords`, location within geographic zone, and timestamp within active period.

T6 — Warm-up attack. `BehavioralArchaeologist` detects rising drift-slope + humanity-slope correlation via σ`(drift_slope × humanity_slope × 20000 − 1.0)`. The sigmoid scoring penalizes coordinated improvement without penalizing natural skill development (which does not produce the characteristic dual-rising slope). Even if a warm-up attack succeeds in accumulating PHG score, Mode 5 enforcement (§7.5.4) can suspend the credential when consecutive critical windows accumulate.

T7 — Distributed bot farm. `FederationBus` (§7.5.6) detects clusters confirmed by $≥$2 independent bridge instances. Even if each shard sees too few devices to form a local cluster, the cross-confirmed hash triggers a `federated_cluster` alert and optional on-chain anchoring.

9.2 Trust Assumptions

Component	Trust Assumption	Failure Mode
CryptoCell-310	Hardware not physically tampered	Key extraction via side-channel
Sensor hardware	Not replaced or physically spoofed	Synthetic data injection (T4)
IoTeX L1	Standard blockchain finality	Reorg reverts verifications
P256 precompile at `0x0100`	Correct implementation	Signature bypass
Bridge service	Honest relay	Withholding/reordering (detected via hash-chain)
Stable EMA track	Initialized from clean sessions	If early sessions are bot-generated, stable track is poisoned

Stable track initialization risk. The L4 biometric anomaly detector computes Mahalanobis distance against a *stable* EMA track updated only on clean NOMINAL sessions. If a device's first 5 sessions are bot-generated but below hard-cheat thresholds, the stable track is initialized from bot biometrics. Subsequent legitimate human play would then trigger false positives while bot sessions appear normal. **Mitigation:** the warmup attack detector (§7.5.5) specifically looks for rising humanity-probability slope early in a device's lifecycle, flagging this bootstrapping attack. The `_STABLE_TRACK_QUARANTINE` invariant comment in `dualshock_integration.py` documents this attack surface explicitly.

9.3 Adversary Capability Boundaries

Adversary Capability	Can Defeat	Cannot Defeat
Host software control only	Kernel anti-cheat, behavioral heuristics	IMU gravity signal (L2), causal coupling (L2B/L2C), active haptic challenge (L6)
Host + USB bus MITM	All software detection	L6 active challenge (requires physical actuator response that software cannot synthesize)
Physical device with known-clean history	Hard cheat detection	Longitudinal drift detection (Mode 5 credential suspension operates across sessions)
Physical device compromise (hardware)	All PITL layers	On-chain chain integrity (hash gaps detectable), credential suspension (longitudinal)

9.4 Limitations

Biometric thresholds calibrated on N=69 sessions, 3 players. The production thresholds (L4 anomaly 7.019, continuity 5.369) are calibrated from 64 included sessions across 3 distinct players. Mode 6 living calibration autonomously refines these thresholds every 6 hours from accumulated NOMINAL records, bounded to ±15% per cycle.

Bridge is a trusted intermediary. The ZK PITL circuit (§7.5.3) constrains the bridge's computation of biometric outputs, but requires the ZK artifact files (`PitlSessionProof.wasm`, `PitlSessionProof_final.zkey`) to be present and the `PITLSessionRegistry` contract to be deployed. Without these, the ZK guarantee is inactive. See §10.2 and §10.3 for the path to full-ZK deployment.

PHGCredential bridge-key is immutable — key compromise enables malicious suspension. The `bridge` address in `PHGCredential.sol` is set at construction time and cannot be changed. If the bridge's signing key is compromised, an attacker can call `suspend()` on any device indefinitely. Until multi-sig or timelock governance is added to the enforcement path, key hygiene for the bridge deployment account is a critical operational security requirement.

No data confidentiality. PoAC records are submitted in plaintext — inference results, action codes, and locations are visible on-chain.

L6 response thresholds are engineering estimates, not empirically calibrated. The L6 Active Physical Challenge-Response classifier (§7.5.2) uses onset and settle thresholds (onset_threshold_ms per profile: 300–450 ms; settle_threshold_ms: 1,500– 2,500 ms) derived from general biomechanical literature rather than measured distributions on real DualShock Edge players. Until a calibration dataset of N$≥$50 real L6 challenge sessions is collected and analyzed, these thresholds carry the same caveat as the pre-N=69 biometric thresholds: plausible but not empirically grounded. False positive and false negative rates for L6 are not yet characterized.

Biometric transplant requires multi-person calibration data (architectural constraint). Inter-person separation has been measured (ratio 0.362) with the N=69 3-player corpus. This ratio is below 1.0, meaning L4 does not currently separate players. Three of the eleven features are structurally zero for all calibration sessions, and tremor FFT requires a wider analysis window than the current 120-frame window at 1000 Hz. Transplant attack detection is not a reliable metric under current conditions — the system is an intra-player anomaly detector, not an inter-player identifier.

Warmup attack graduation is a fundamental limitation of session-scoped behavioral detection. A sufficiently slow warmup attacker who interpolates metrics to human baseline over many sessions will eventually graduate past the detection boundary. Session-scoped detectors (BehavioralArchaeologist, L4 biometric, L5 temporal) observe behavior within a window; they cannot detect an attacker who consistently stays within the human-behavior region during each individual session. The defense for this attack class is multi-session temporal analysis (Mode 5 credential suspension), not within-session PITL scoring.

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10. Discussion and Future Work

10.1 Real-Hardware Adversarial Validation

Addressed (2026-03-07). The adversarial validation gap identified in earlier versions of this paper has been closed for the three-player case:

- **N=69 biometric calibration** — 69 DualShock Edge sessions captured at 1000 Hz across 3 distinct players; L4/L5 thresholds derived empirically via `scripts/threshold_calibrator.py`
- **6-attack adversarial suite** — 55 sessions generated by deterministic transforms of real captures; full L2/L4/L5 pipeline validated (see Table 5, §8.3)

- **Detection rates:** injection 100%, macro 100%, quant-masked 100%, warmup 60%
- **Human false positive rate:** 2.9% (2/69 — within 3σ statistical expectation)
- **Stationary control baseline** — 30s idle controller capture confirming IMU noise floor (gyro std 1.3–1.5 LSB) and 14,000× injection detection margin
- **L6 Active Physical Challenge-Response implemented** — 8-profile trigger challenge library, async trigger driver, motor-response curve analyzer, Attack G adversarial unit tests. Human response baseline calibration is the remaining open item (§10.6).
- **PHGCredential auto-expiry fix** — `isActive()` now honors `suspendedUntil` timestamp; `suspend()` allows re-suspension after auto-expiry without requiring `reinstate()`. 354 Hardhat tests passing (+2 auto-expiry tests). Also: CEI pattern in `PoACVerifier._verifyInternal()` confirmed correct and documented with comment.

Remaining work: - Professional bot software (aimbot trajectories, macro tools) as labeled adversarial data - Skill-tier diversity (Bronze through Diamond) to validate false positive rate across play styles rather than a single player's sessions - Recapture sessions after Phase 17 so touchpad features (now populated) contribute to inter-person separation

10.2 Toward Full-ZK PoAC

As zkML tooling matures, PoAC could incorporate succinct proofs of correct inference execution per cognition cycle, upgrading model attestation from "the device claims to have used model $\mathcal{M}$" to "the device provably executed $\mathcal{M}$ on input x." The primary barrier is prover time on embedded hardware. Application-specific circuits for fixed TinyML architectures and hardware ZK accelerators could make this feasible within 3–5 years.

10.3 Multi-Instance Trusted Setup

The current ZK ceremony (`contracts/scripts/run-ceremony.js`) is a single-contributor development ceremony. A production deployment requires a multi-party MPC ceremony (Hermez Perpetual Powers of Tau or equivalent) to ensure that no single participant can reconstruct the toxic waste.

10.4 Full Covariance Biometric Fingerprinting

The current L4 classifier uses a diagonal covariance assumption (7 independent variances). A full 7×7 covariance matrix would capture cross-feature correlations (e.g., trigger onset velocity is correlated with grip asymmetry for a given player) and improve both sensitivity and specificity. The `controller/tinyml_biometric_fusion.py` TODO comment documents this as the next algorithmic improvement.

10.5 Formal Verification

The PoAC chain integrity properties (linkage, monotonicity, non-repudiation) are amenable to formal verification in TLA+ or Isabelle/HOL. Machine-checked proofs would strengthen confidence for safety-critical esports deployments.

10.6 L6 Human Response Baseline Calibration

L6 Active Physical Challenge-Response (§7.5.2) uses onset/settle thresholds and classification weights derived from general biomechanical priors. The next hardware validation milestone is a calibration study analogous to the N=69 biometric calibration (§8.4) but targeting the challenge-response dimension:

1. **Capture N$≥$50 challenge sessions** — run `L6_CHALLENGES_ENABLED=true` with a live DualShock Edge, dispatching all 8 profiles across gameplay sessions
2. **Characterize human distributions** — fit onset_ms, settle_ms, peak_delta, grip_variance distributions per profile; derive profile-specific thresholds
3. **Attack G ground truth** — collect replay-injection sessions with L6 active; measure false negative rate under Attack G
4. **Update `CHALLENGE_PROFILES`** — replace engineering-estimate thresholds with measured mean ± 3σ values via a `scripts/l6_threshold_calibrator.py` script analogous to `scripts/threshold_calibrator.py`

Until this calibration is performed, L6 is recommended as a supplementary layer only (`L6_CHALLENGES_ENABLED` default) and should not be used as a primary gating signal for tournament qualification.

10.7 Multi-Person Biometric Calibration (Transplant Attack Validation) — Updated

The N=69 calibration corpus now spans 3 distinct players (hw_005–hw_073). Inter-person Mahalanobis separation has been computed (separation ratio 0.362 — see docs/interperson-separation-analysis.md). The result is honest: L4 does not currently separate players. Three of eleven features are structurally zero (touchpad, trigger resistance, touch position) and two (tremor FFT) require wider FFT windows than the current 120-frame window at 1000 Hz. The next milestones are: (1) recapture sessions after Phase 17 (touchpad features now populated), (2) widen the FFT analysis window to $≥$512 frames, (3) validate whether the corrected feature set achieves inter-person separation.

---

11. Conclusion

We have presented VAPI — a system providing verifiable provenance for controller input composed with physics-backed liveness detection. PoAC's 228-byte chained evidence record captures the complete cognitive context: what was sensed, what model produced the inference, what the agent's accumulated world model contained at decision time, and what action was taken — all committed, signed, hash-chained, and anchored on a public blockchain. Any third party can verify the origin, ordering, and integrity of a session's evidence log without trusting the bridge or any other intermediary.

The DualShock Edge's motorized adaptive trigger surface is the key physical primitive: a PoAC chain anchored to resistance dynamics, six-axis IMU, and stick kinematics cannot be reproduced by software injection. The nine-level PITL stack exploits signals grounded in physics: IMU gravity (absent in injected data), IMU-button causal latency (present only when a physical hand precedes each press), stick-IMU temporal cross-correlation (present only when a physical hand couples stick movement to body sway), biometric kinematic fingerprinting (individual-specific across 11 features), temporal rhythm analysis (human timing variance cannot be faked with constant-interval scripting), and active haptic challenge-response (onset latency, grip variance, and settling behavior are involuntary biomechanics that software cannot sense or replicate).

The PHG humanity credential is a *living* proof: earned through sustained clean behavior, weighted by biometric quality, portable across key rotations through biometric continuity, and provisionally suspended when retrospective memory accumulates evidence of sustained adversarial behavior. Mode 6 living calibration (Phase 38) adds a new class of self-improvement: the system evolves its own detection thresholds from verified session data every 6 hours, with per-player profiles that tighten detection for known players without ever loosening it — a credential that improves with every verified session.

The complete system (~220 files, ~1,289 automated tests including 28 on physical hardware) demonstrates the concept is implementable today with existing gaming controller hardware and existing blockchain infrastructure. Thirteen contracts are deployed on IoTeX testnet. Live hardware validation confirms the foundational physical signal claims: USB polling at 1002 Hz, gyro noise 14,000× above the stationary-control baseline, 278,239 LSB^2 accel variance from natural hand micro-tremor, and zero report-counter violations across 200 consecutive reports.

Honest limits: L4 is an anomaly detector, not a player identifier, with the current feature set and single-game corpus. The ZK proof does not bind inference codes on-chain. L6 human-response thresholds lack empirical calibration. The bridge remains operationally trusted despite ZK constraints.

VAPI opens a design space where provenance and physical plausibility of gaming sessions are verifiable — not assumed. The infrastructure to extend, recalibrate, and strengthen every layer of this system ships with this release.

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Appendix A: DePIN Economic Layer

VAPI's DePIN layer demonstrates that PoAC extends naturally to non-gaming sensor domains without protocol changes. An IoTeX Pebble Tracker (nRF9160 SiP, ARM Cortex-M33 @ 64 MHz, CryptoCell-310) performs autonomous bounty participation: it discovers active environmental monitoring tasks on-chain, evaluates fit against available time and energy budget, accepts the best-fit task, and preempts lower-value tasks when higher-value opportunities arrive mid-session. Each economic decision — accept, continue, preempt — generates a PoAC record committed with the same 228-byte wire format, device-key ECDSA-P256 signature, and on-chain anchoring used for gaming sessions. The DePIN layer thus validates two design claims simultaneously: (1) PoAC is device-agnostic — the same verification mechanism serves both gaming controller inputs and environmental telemetry; and (2) economic decision-making (not just raw sensing) is attestable within the PoAC framework.

Bounty evaluation. Available tasks form a knapsack instance over device capacity. Each task carries a reward r, expected duration d, and geographic zone z. The greedy evaluator selects the highest reward-per-second task fitting within the remaining session window and preempts the current task when a new opportunity offers $\geq 1.5\times$ the active reward rate. Over 1,000 synthetic scenarios, this policy achieves a median 97.1% of optimal total reward (mean 94.2%, worst-case 81.3%) in 0.14 ms mean decision time on the nRF9160 (cycle-accurate emulation). The preemption threshold fires in 12.7% of scenarios. All DePIN evaluation figures are simulation-derived; real-hardware Pebble Tracker validation is future work.

Appendix B: BridgeAgent — Complete Tool Catalogue and Interface Specification

`BridgeAgent` (`claude-sonnet-4-6`, `bridge/vapi_bridge/bridge_agent.py`) provides LLM-powered operator intelligence through 17 deterministic tool bindings. All tools are read-only against the SQLite store and on-chain state; no tool mutates bridge state.

B.1 Tool Catalogue

Tool	Returns
`get_player_profile`	PHG score, checkpoint count, risk label, credential status

Tool	Returns
`get_leaderboard`	Top-N devices by confirmed PHG score
`get_leaderboard_rank`	Single device rank within leaderboard
`run_pitl_calibration`	L4/L5 threshold suggestions from live DB distribution
`get_continuity_chain`	Session continuity attestation history for a device
`get_recent_records`	Last N PoAC records with PITL inference codes
`get_startup_diagnostics`	ZK artifact presence, contract addresses, feature flags
`get_phg_checkpoints`	Full PHG checkpoint chain (up to limit 50)
`check_eligibility`	Tournament eligibility: PHG score + credential active
`get_pitl_proof`	Latest ZK PITL session proof row
`get_behavioral_report`	`BehavioralArchaeologist` analysis: drift slope, warmup, burst
`get_network_clusters`	`NetworkCorrelationDetector` clusters filtered by min suspicion
`get_federation_status`	Peer count, cross-confirmed clusters, federation enabled
`query_digest`	`InsightSynthesizer` digest for 24h / 7d / 30d window
`get_detection_policy`	Active L4 threshold multiplier and basis risk label
`get_credential_status`	Evidence chain: biometric label → suspension state → reinstatement conditions
`get_calibration_status`	Global L4 thresholds, per-player profiles, recent threshold evolution, next Mode 6 cycle timing

B.2 Streaming Interface

`GET /operator/agent/stream` (API-key gated, rate-limited to 60 req/min) returns Server-Sent Events with the following typed event schema:

`type` field	Payload
`text_delta`	`{text: str}` — incremental reasoning token
`tool_start`	`{tool_name: str, inputs: dict}` — visible tool invocation
`tool_result`	`{tool_name: str, result: any}` — tool return value
`done`	`{session_id: str, tools_used: list[str]}` — completion summary
`error`	`{message: str}` — non-fatal error within stream

The 5-round agentic loop ensures tools can chain (e.g., `get_player_profile` → `get_behavioral_report` → `get_credential_status` in a single natural-language query).

B.3 Autonomous Reaction

`BridgeAgent.react(event: dict)` handles `BIOMETRIC_ANOMALY` (0x30) and `TEMPORAL_ANOMALY` (0x2B) events autonomously. The method: - Uses an internal session namespace `__react_{device_id[:8]}` (isolated from operator sessions) - Never raises — all exceptions are caught and returned as error dicts - Persists each reaction to `protocol_insights` table as an auditable record - Returns `{alert: str, severity: str, tools_used: list, device_id: str, inference: int}`

B.4 Session Persistence and History Compression

Session history is stored in the `agent_sessions` SQLite table (schema: `session_id TEXT PK, history_json TEXT, created_at REAL, updated_at REAL`) and survives bridge restarts. When history exceeds `AGENT_MAX_HISTORY_BEFORE_COMPRESS` (default 60, configurable), the compressed portion is replaced with a summary entry:

```
[System: N prior messages compressed. Tools used: tool×count.
 Continue from the 20 most recent messages below.]
```

The tool-use inventory is extracted from the compressed messages before replacement, preserving operator context across long investigation sessions.